Zsigmondy Congruence-Return Towers and the Global Determinant-Owner Obstruction

Route-A structural certificate C179

Abstract

For every coprime pair $a > b \geq 1$, multiplication by ab^{-1} on $(\mathbb{Z}/N\mathbb{Z})^\times$ defines an arithmetic dynamics without fitted data. A prime p is primitive for $a^n - b^n$ exactly when the marked point 1 first returns at time n on the p-fiber. Classical Zsigmondy existence supplies such a prime outside its two exact exception forms; that theorem is attributed, not claimed new. If $e = v_p(a^n - b^n)$, the least return on every p^k-fiber is $np^{\max(0,k-e)}$. Every admissible finite fiber has a complete cycle, fixed-point, zeta, determinant, and reversor ledger. Globally, however, the disjoint union has fixed ledger $a^n - b^n$ and zeta $(1-bz)/(1-az)$, whereas the profinite inverse-limit translation has no positive-time fixed points and zeta 1. Thus finite fibers alone do not select a unique global determinant owner.

Keywords: primitive divisor; first return; congruence dynamics; multiplicative order; Zsigmondy theorem.

中文摘要

对互素整数 $a > b \geq 1$，在有限单位群 $(\mathbb{Z}/N\mathbb{Z})^\times$ 上乘以 ab^{-1} 给出一个无需拟合数据的算术动力系统。本文证明：素数 p 是 $a^n - b^n$ 的本原素因子，当且仅当标记点 1 在模 p 纤维上首次于时刻 n 返回。经典的 Zsigmondy 定理给出除两个精确例外类型之外的存在性； 本文明确归属该经典结果，并不声称其为新定理。若 $e = v_p(a^n - b^n)$，模 p^k 纤维的最小返回时间为 $np^{\max(0,k-e)}$；每个允许的有限纤维都有完整的 循环、不动点、ζ、行列式与反演账本。然而，两种自然全局化并不一致： 不交并的第 n 步不动点数为 $a^n - b^n$，而逆极限平移在所有正时间均无不动点。因此，有限纤维本身不能选出唯一的全局行列式载体。

1 Frozen arithmetic dynamics

Fix coprime integers $a > b \geq 1$. For every $N \geq 2$ with $(N, ab) = 1$, set

$$U_N = (\mathbb{Z}/N\mathbb{Z})^\times, \qquad q_N = ab^{-1} \pmod N, \qquad R_N(x) = q_N x,$$

and mark $1 \in U_N$. One multiplication is one discrete source step. No logarithmic prime roof, prime weight, target table, or fitted parameter is introduced.

2 Primitive-return theorem

Theorem. For every prime p and $n \geq 2$, the following are equivalent:

1. $p \mid a^n - b^n$ and $p \nmid a^m - b^m$ for $1 \leq m < n$;
2. $p \nmid ab$ and $\mathrm{ord}_p(ab^{-1}) = n$;
3. 1 has least positive return n under R_p.

Moreover, such a prime exists except when $(a, b, n) = (2, 1, 6)$, or when $n = 2$ and $a + b$ is a power of two.

Proof. A prime divisor of $a^n - b^n$ cannot divide ab. Thus $a^m \equiv b^m \pmod p$ is equivalent to $(ab^{-1})^m \equiv 1 \pmod p$. Excluding all earlier m is exactly the definition of multiplicative order, and the orbit of the mark is $1, q_p, q_p^2, \ldots$. The existence and exception sentence is the classical Zsigmondy theorem [1]; no new proof or strengthening of that theorem is claimed. □

Every primitive return prime for $n \geq 2$ is odd. Indeed, if a, b are odd, then $2 \mid a - b$ already at time one; if they have opposite parity, every $a^n - b^n$ is odd.

3 Exact lift through every prime power

Let p be primitive at n, and set $e = v_p(a^n - b^n)$. Then p is odd and $n = \mathrm{ord}_p(ab^{-1}) \mid p - 1$. For every $k \geq 1$,

$$\boxed{\mathrm{ord}_{p^k}(ab^{-1}) = np^{\max(0,k-e)}}. \tag{1}$$

Indeed, an exponent returning modulo p^k must be nr. Since b is a p-adic unit, odd-prime lifting of the exponent gives

$$v_p((ab^{-1})^{nr} - 1) = e + v_p(r).$$

The least eligible r is $p^{\max(0,k-e)}$. Thus every orbit on the p^k-fiber has the length in (1), and the number of cycles is $\varphi(p^k)/(np^{\max(0,k-e)})$.

4 Complete dynamics on every finite fiber

For any admissible N, write $L_N = \mathrm{ord}_N(q_N)$. Translation returns a point x exactly when $q_N^t = 1$, independently of x. Hence U_N is $\varphi(N)/L_N$ cycles, each of length L_N, and

$$\#\,\mathrm{Fix}(R_N^t) = \begin{cases} \varphi(N), & L_N \mid t, \\ 0, & L_N \nmid t, \end{cases} \qquad \zeta_N(z) = (1 - z^{L_N})^{-\varphi(N)/L_N}. \tag{2}$$

For the finite permutation Koopman matrix $\mathcal{U}_N$, using the fixed-count convention of Artin and Mazur [3],

$$\det(I - z\mathcal{U}_N) = (1 - z^{L_N})^{\varphi(N)/L_N}. \tag{3}$$

Inversion $J_N(x) = x^{-1}$ satisfies $J_N R_N J_N = R_N^{-1}$ because $(q_N x^{-1})^{-1} = x q_N^{-1}$.

5 Two globalizations and owner nonselection

Adjoin the singleton U_1 and act fiberwise on $\mathcal{D}_{a,b} = \bigsqcup_{(N,ab)=1} U_N$. The N-fiber is fixed at time n exactly when $N \mid a^n - b^n$. Every such divisor is admissible, and the divisor–totient identity gives

$$\#\,\mathrm{Fix}(R^n) = \sum_{N \mid a^n - b^n} \varphi(N) = a^n - b^n. \tag{4}$$

Consequently,

$$\zeta_{\mathcal{D}}(z) = \exp\left(\sum_{n \ge 1} \frac{a^n - b^n}{n} z^n \right) = \frac{1 - bz}{1 - az}, \qquad C_n = \frac{1}{n} \sum_{d \mid n} \mu(n/d)(a^d - b^d), \tag{5}$$

where C_n counts primitive cycles of length n.

Now take the inverse limit $\widehat{U}^{(ab)} = \varprojlim_{(N,ab)=1} U_N$ and translate by the compatible element $q = a/b$. A time-n fixed point would force $q^n = 1$ in every finite quotient. Choosing a prime outside the finite divisor set of $ab(a^n - b^n)$ contradicts this. Hence

$$\mathrm{Fix}(R^n \mid \widehat{U}^{(ab)}) = \varnothing \quad (n \ge 1), \qquad \zeta_{\widehat{U}}(z) = 1. \tag{6}$$

Equations (4) and (6) are incompatible fixed ledgers for two source-natural globalizations. Therefore the finite congruence fibers do not select one global periodic-orbit determinant owner. This is a no-go for uniqueness without extra structure, not an absolute claim that every enlarged owner is impossible.

Same-clock operator corollary. Counting measure makes each finite permutation Koopman operator unitary, and normalized counting measures are compatible with the inverse-limit Haar probability measure. Translation by q therefore induces a canonical unitary Koopman operator on $L^2(\widehat{U}^{(ab)})$. Fiberwise inversion and profinite inversion reverse these operators on the unchanged discrete clock. This supplies the natural A4 lift; it does not choose between (4) and (6), nor does it create a target determinant.

6 Exact audit and Route-A decision

The ledger has 567 primitive-divisor, 2,080 prime-power, 1,650 finite-fiber, and 630 globalization rows. Its seven no-return rows are exactly the classical exceptions. A producer-independent checker passes 320,291 assertions; SymPy passes 6,674 exact checks; byte replay is exact; 64 repaired-hash mutations and one stale-hash mutation are rejected. These are regression sentinels; (1)–(6) have no cutoff.

Gate	Verdict	Exact boundary
A0	`WEAK`	primes are intrinsic first-return moduli, not one global owner
A1	`WEAK`	every finite fiber is exact; global ledgers disagree
A2	`FAIL`	no target divisor or frozen target validation protocol
A3	`FAIL`	no target analytic law or Weil compression
A4	`NATURAL`	finite permutation and profinite Haar Koopman lifts

The exact tuple is

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(A0_WEAK_ARITHMETIC_RELATION,A1_WEAK,A2_FAIL,A3_FAIL,
A4_NATURAL_QUANTIZATION).
```

Overall: `ROUTE_A_EXPLORATORY`; Route B is false. A0 is weak, not passed: no logarithmic prime clock or selected global prime-orbit owner exists. A4 records canonical same-clock source unitaries and repairs none of A0–A3.

Attribution boundary. Zsigmondy's existence theorem is external. Birkhoff–Vandiver [2] supplies historical divisor context, while Silverman's dynamical Zsigmondy work [4] supplies modern context only. Neither is presented as proof of the package's lift, finite-fiber, or globalization theorems.

Scope and limitations. No target zero or prime table, local arithmetic factor, root number, automorphy input, prime-weighted global product, logarithmic prime roof, or Route-B authorization enters this note. Scope literal: `NO_BAD_EULER_OR_ROOT_NUMBER`. A primitive prime labels a finite return fiber; it is not identified with an isolated prime-labeled orbit inside one global phase space. The rational function in (5) is an unweighted source zeta, not an arithmetic local factor or a product indexed by primes. The profinite result rules out fixed points for that natural inverse-limit translation only. It does not rule out a future enlarged model supplied with an independently justified owner-selection principle.

Declarations

Data and code. Package-local exact evidence and deterministic code accompany the paper. **Ethics.** No human, animal, clinical, personal, private, or sensitive data are used; approval is not applicable. **CRediT.** This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing; AI systems are not authors. **Funding.** No external funding. **Conflicts.** None known. **AI use.** An AI coding assistant supported drafting and exact-code development; it was not an external reviewer or an independent error process.

References

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